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Paper 1: More Than a Question of Correlation: Characterization of the Evidentiary Basis f

Bibliographic Information

- **Title**: More Than a Question of Correlation: Characterization of the Evidentiary Basis for Biomarker Surrogates Used in European Marketing Authorizations
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Paper Type: GENERIC

Quick Take

This study systematically identified 14 novel biomarker surrogates used as primary endpoints in EU marketing authorizations for cardiovascular and metabolic medicines (2008–2023). For 13 of 14 (used in orphan drug trials), correlation with clinical outcomes was not well-established in the literature; only 4 had dedicated validation studies. Regulators accepted these surrogates based on totality of evidence and high unmet need, highlighting a gap between evidentiary standards and regulatory practice.

Executive Summary

The authors examined European Public Assessment Reports (EPARs) to identify biomarkers first used as surrogate primary endpoints in pivotal trials supporting EU marketing authorizations for cardiovascular and metabolic medicines. Of 35 biomarkers identified, 14 were novel surrogates, all but one for orphan-designated medicines. Systematic literature searches revealed that for most biomarkers, evidence of correlation with the intended clinical outcome was limited to low-hierarchy observational studies; only 5 publications (4.3%) were meta-analyses. Justification for surrogate choice was rarely explicit in public EPARs, though pre-assessment regulatory interactions (scientific advice or protocol assistance) occurred for most. The findings underscore that regulators accept surrogate endpoints in rare diseases despite limited validation, relying on mechanistic plausibility and the totality of evidence.

Scientific Context & Motivation

Biomarkers can serve as surrogate endpoints when direct measurement of clinical outcomes (e.g., survival, organ failure) is impractical due to long latency, low event rates, or small populations—common in rare diseases. However, robust validation requires demonstration of strong correlation between biomarker change and clinical outcome, ideally through meta-analyses of randomized trials. Prior work in oncology has shown that many accepted surrogates lack high-level evidence.

This study addresses the knowledge gap for cardiovascular and metabolic indications in the EU, focusing on the evidentiary basis available at the time of first regulatory acceptance.

⚡ Methodological Snapshot

The study had two phases. First, novel biomarker surrogates were identified by screening EPARs for all cardiovascular (ATC-C) and metabolic (ATC-A) medicines authorized in the EU between 2008 and 2023. A biomarker was considered 'novel' if it had not been used as a primary surrogate endpoint in any prior EU authorization for the same indication, verified via a database of all EPARs since 1995. Fourteen such surrogates were identified. Second, for each surrogate, systematic literature searches were conducted in PubMed and Embase to identify publications that investigated or discussed the correlation between the biomarker and the clinical outcome it was meant to replace. Searches combined biomarker and indication terms; if >500 hits, an additional surrogate-specific concept was added. Abstracts were screened by one reviewer with $\geq 10\%$ dual screening; discrepancies were resolved by consensus. Included publications were categorized using a three-level hierarchy of evidence (meta-analyses = level 1, observational/review = level 2, case studies = level 3). Data on study design, funding, citation count, and author groups were extracted. The analysis was descriptive.

📋 Detailed Analysis

The study's core contribution is a systematic, regulatory-focused mapping of how novel biomarker surrogates enter clinical practice via marketing authorizations. The finding that 13 of 14 surrogates were for orphan medicines underscores the pragmatic reality: in rare diseases, traditional clinical endpoints are often infeasible, and regulators accept surrogates with limited validation. However, the near absence of high-level evidence (only 5 meta-analyses across all 14 surrogates) and the lack of explicit justification in public EPARs (only 1 of 14 with references) reveal a transparency gap. The authors rightly note that pre-assessment regulatory interactions (scientific advice/protocol assistance) likely contained detailed rationale, but this remains confidential. This limits the ability of other developers to learn from precedent. The study also highlights that most included publications were published within 5 years of first authorization, suggesting that evidence generation often coincides with—rather than precedes—regulatory use. The use of the Ciani hierarchy is appropriate but may undervalue mechanistic or biological plausibility arguments, which are particularly relevant for rare diseases where randomized trial data are scarce. The study's focus on cardiovascular and metabolic indications is a strength, but the findings likely generalize to other therapeutic areas with high orphan drug activity. A notable omission is the lack of assessment of post-authorization validation: did these surrogates later prove to predict clinical outcomes? Future work should track the long-term performance of these surrogates.

Domain Context

This paper sits at the intersection of regulatory science, clinical pharmacology, and translational medicine. In pharmacometrics, surrogate endpoints are often modeled using longitudinal biomarkers (e.g., eGFR slope, tumor size) to predict long-term outcomes. The paper's findings challenge the assumption that such surrogates are rigorously validated before regulatory acceptance. For the pharmacometrics community, this underscores the importance of quantitative approaches (e.g., disease progression models, meta-analytic surrogacy evaluation) to strengthen the evidentiary basis for surrogates. The paper also relates to the EMA's qualification procedure, which provides a formal pathway for biomarker acceptance—yet none of the identified surrogates had been qualified. This suggests that the qualification procedure may be underutilized or too burdensome for rare diseases. The call for better reporting aligns with ongoing initiatives like the CONSORT-Surrogate Extension and the FDA's surrogate endpoint table. For drug developers, the key takeaway is that early regulatory dialogue is critical, and that mechanistic plausibility combined with supportive (even if non-randomized) data can suffice for orphan indications.

□ Key Findings

Fourteen novel biomarker surrogates were identified, including serum bile acids (Alagille syndrome, PFIC), eGFR slope (Fabry disease), total kidney volume (ADPKD), urinary oxalate (primary hyperoxaluria), and others. All but one (decrease in eGFR $\geq 40\%$ for CKD) were used in orphan-designated medicines. Systematic literature searches found that for 10 of 14 biomarkers, fewer than 10 publications explicitly investigated correlation with the clinical outcome. Only 5 publications (4.3%) were meta-analyses (hierarchy level 1); the vast majority (94%) were observational studies or reviews. Justification for surrogate choice was provided in only 6 of 13 EPARs, and only one (lumasiran) included explicit literature references. Pre-assessment regulatory interactions occurred for 9 of 13 medicines, but detailed rationale remained confidential.

Strengths & Limitations

Strengths

- Systematic, reproducible methodology for identifying novel biomarker surrogates from regulatory documents
- Comprehensive literature search across two major databases with dual screening and validation
- Use of a published hierarchy of evidence for surrogate endpoint validity to categorize studies
- Focus on a 15-year period providing a contemporary view of regulatory practice
- Inclusion of both cardiovascular and metabolic indications, broadening relevance beyond oncology

Limitations (Acknowledged by Authors)

- Strict abstract screening may have excluded relevant evidence not explicitly mentioned in abstracts
- Addition of a search concept for biomarkers with >500 hits may have missed relevant publications

- Only cardiovascular and metabolic indications were studied; findings may not generalize to other therapeutic areas
- Influence of secondary endpoints on regulatory acceptance was not assessed

Limitations (Expert Review)
- The study did not assess the quality or strength of correlation evidence within included publications (e.g., effect sizes, confidence intervals)
- Reliance on EPARs as the sole source for identifying novel surrogates may miss biomarkers used in post-authorization studies or extensions
- The hierarchy of evidence used (Ciani et al.) may not fully capture all forms of validation (e.g., mechanistic or biological plausibility arguments)
- No assessment of whether the same biomarkers were subsequently validated after initial acceptance

Generalizability
The findings are directly applicable to cardiovascular and metabolic medicines in the EU regulatory context. The predominance of orphan indications suggests limited generalizability to non-orphan settings. However, the pattern of accepting surrogates with limited validation may extend to other therapeutic areas where rare diseases are common (e.g., neurology, hematology).

Figures & Tables
- **Figure 1**: Schematic representation of the systematic literature review methodology for each novel biomarker surrogate.
  - *Significance*: Clarifies the stepwise search and screening process, including the addition of a surrogate-specific search concept when initial hits exceeded 500.
- **Figure 2**: Flowchart showing selection of novel biomarker surrogates from 35 initial biomarkers identified in EPARs.
  - *Significance*: Documents the reproducible process of identifying novel surrogates and shows that 14 of 35 were considered novel.
- **Figure 3**: Bar chart showing number of publications identified, screened, and included for each of the 14 biomarker surrogates.
  - *Significance*: Visually demonstrates the wide variation in available literature across biomarkers and the low proportion of publications that met inclusion criteria.
- **Table 2**: Characteristics of the 14 novel biomarker surrogates, including indication, clinical outcome replaced, medicine, year, ATC code, authorization type, and number of main efficacy studies.
  - *Significance*: Core dataset showing that 13 of 14 surrogates were for orphan medicines and that most were used in single pivotal studies.
- **Table 3**: Justification for choice of primary endpoint as presented in EPARs, including quotes and references.
  - *Significance*: Highlights the limited and often non-referenced justification provided in public documents; only one EPAR included explicit literature references.

- **Table 4**: Characteristics of included publications for each biomarker surrogate: number, author groups, origin, funding, publication date, citations, study design, and hierarchy level.
  - *Significance*: Demonstrates that most evidence is from low-hierarchy observational studies, with only 5 meta-analyses across all biomarkers.

Supplementary Materials

Data S1 contains detailed search strategies for each biomarker surrogate in PubMed and Embase. Table S1 defines the hierarchy of evidence for surrogate endpoint validity. Table S2 lists all 35 biomarkers initially identified. Table S3 provides the full extraction dataset for included publications. Table S4 defines categories of justification used in Table 3. Figure S1 shows the complete selection process per biomarker surrogate.

Future Directions

Future research should investigate how secondary endpoints influence regulatory acceptance of surrogate primary endpoints, and whether biomarkers accepted as surrogates are later validated in post-marketing studies. The CONSORT-Surrogate Extension should be promoted to improve trial reporting. The EMA could publish a publicly accessible list of accepted surrogate endpoints with supporting rationale, similar to the FDA's table of surrogate endpoints. Studies in other therapeutic areas (e.g., oncology, neurology) would complement these findings.

Expert Commentary

This paper provides a sobering reality check for pharmacometricians and translational scientists: the evidentiary bar for surrogate endpoint acceptance in rare diseases is often lower than textbook ideals. While mechanistic plausibility and regulatory flexibility are understandable given unmet medical need, the lack of high-level validation (meta-analyses) raises questions about the reliability of benefit-risk assessments. The finding that only one EPAR explicitly referenced literature is concerning. As a community, we should advocate for more rigorous pre-approval validation frameworks—perhaps leveraging quantitative systems pharmacology models or Bayesian meta-analyses—to strengthen the link between biomarkers and clinical outcomes. The PRIME-CKD consortium's work is a step in the right direction.

Bottom Line

Regulators accept novel biomarker surrogates in orphan drug approvals even when high-level evidence of correlation with clinical outcomes is lacking, relying on mechanistic plausibility and totality of evidence. Drug developers should engage early with regulators (scientific advice/ protocol assistance) and document surrogate rationale transparently. The field needs clearer guidance on evidentiary standards for surrogate endpoint validation, and public reporting (e.g., in EPARs) should be more explicit to foster innovation and regulatory learning.

Figures

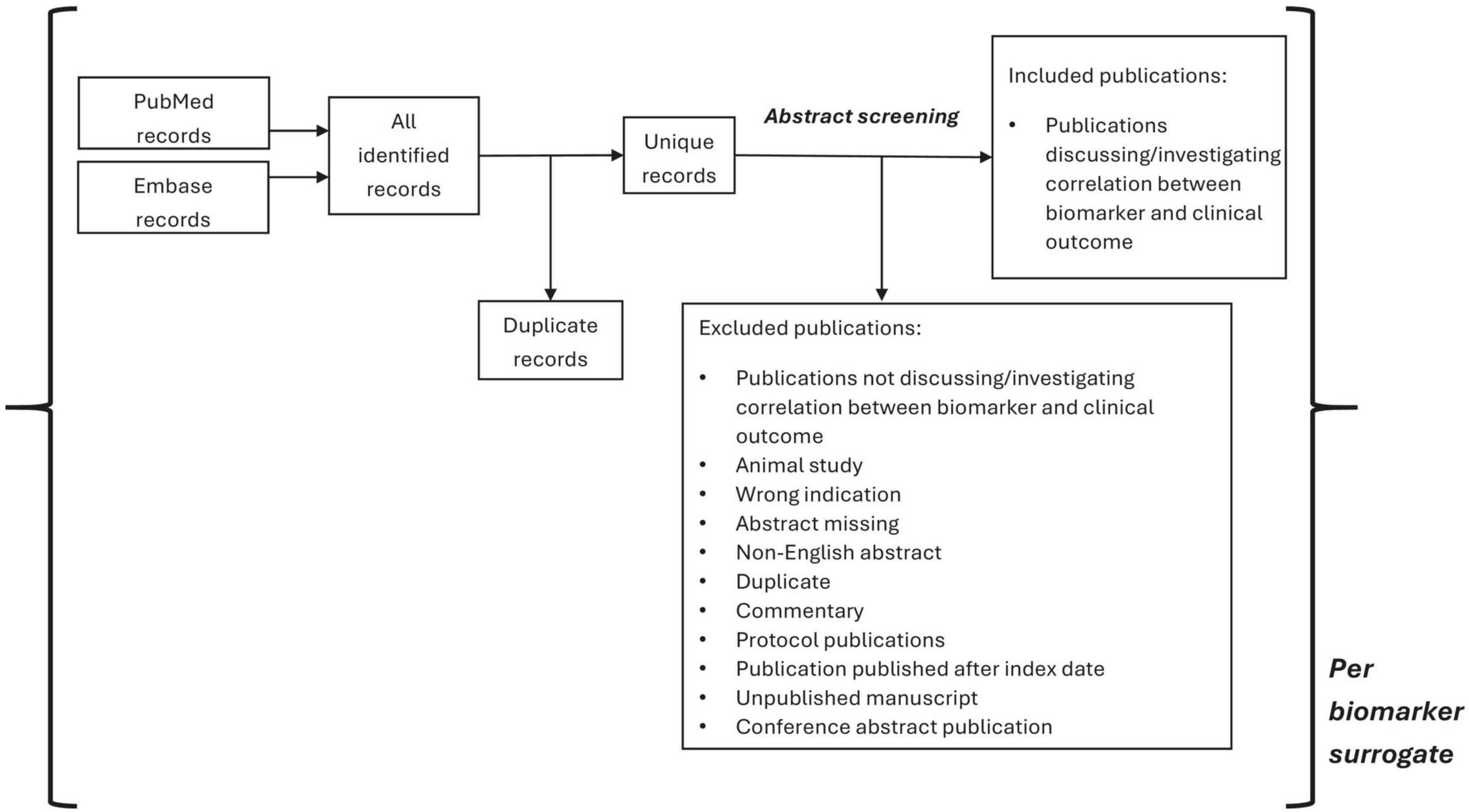

Figure 1: Schematic representation of the utilized methodology for the literature review for each identified novel biomarker surrogate.

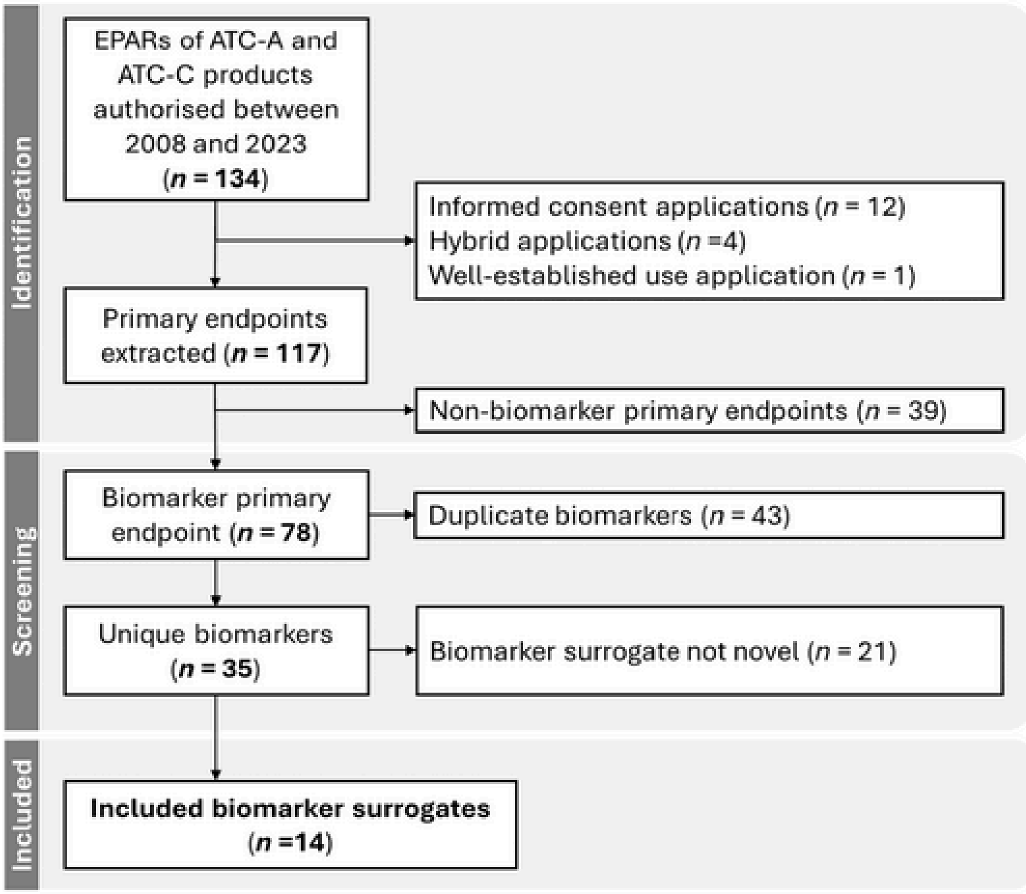

Figure 2: Flowchart for the selection of novel biomarker surrogates. EPARs were selected from the table of all EPARs from human and veterinary medicines derived from the E

Figure 3: Overview of the identified, screened, and included publications for each biomarker surrogate. The gray bars represent the initially identified publications prior